

Convertible Bonds

A Guide

January 2006

Christopher Davenport

+44 20-7986 0233



Introduction

This guide looks mainly at the Europe convertibles market. Much of the content is applicable to convertibles markets in general, however. We outline practical and theoretical aspects of the market in eight sections. There is also a glossary of terms.

The basics

Using a hypothetical example, the key features of a typical convertible issue are introduced and their significance explained.

Convertibles bonds: equity investor perspective

From one perspective the convertible bond resembles equity with an element of downside protection and different income characteristics.

Convertibles bonds: bond investor perspective

A convertible bond combines the characteristics of a bond with a degree of equity participation.

Convertible bonds: hedge fund perspective

Hedge funds typically neutralise exposure to one or more of the variables that influence a convertible's value.

Convertible bonds: issuer perspective

Numerous theoretical and practical considerations face the potential issuer of equity-linked paper.

Convertible structures

We look at a number of structures including the incongruously named 'mandatory convertible'.

Pitfalls and protections

Prospectus issues are discussed.

Pricing models

Aspects of convertibles modelling are introduced.



Table of Contents

Basics.....	4
Convertible Bonds: Equity Investor Perspective	24
Convertible Bonds: Bond Investor Perspective.....	27
Convertible Bonds: Hedge Fund Perspective	29
Convertible Bonds: Issuer Perspective	41
Convertible Structures	48
Pitfalls and Protection	59
Convertible Pricing Models.....	66
Appendix.....	70
Glossary	78

Index of Figures

Figure 1. Chimera 4% 2011 — Convertible Data at Issue.....	6
Figure 2. Chimera — Ordinary Share Data	6
Figure 3. Chimera 4% 2011 — Convertible Measures at Issue.....	6
Figure 4. Premium Against Share Price.....	18
Figure 5. Payoff Profile of Convertible Hedged on a Static Delta.....	26
Figure 6. Chimera Convertible 4% 2011 Long Convertible Position.....	30
Figure 7. The Convertible Structure Spectrum	49
Figure 8. Zero Coupon Bond with Premium Redemption Structure	50
Figure 9. Vanilla Convertible Structure.....	51
Figure 10. Rate of Return to Maturity.....	52
Figure 11. DEC's: number and value of shares received	55
Figure 12. DEC's Payoff Chart.....	55
Figure 13. Convertible Price Against Parity	72
Figure 14. Delta Against Parity.....	73
Figure 15. Gamma Against Parity.....	74
Figure 16. Vega Against Parity.....	75
Figure 17. Rho Against Parity.....	76

Basics

- **Convertible bonds as hybrid securities**
- **The 'vanilla' structure**
- **Terminology**
- **Convertibles as derivatives**
- **Valuation stimuli**

Convertible bonds as hybrid securities

Convertibles are corporate bonds that may be exchanged at the option of the holder for a fixed number of ordinary shares. At maturity, a convertible is worth the greater of its cash redemption value or the market value of the shares into which it carries rights of conversion.

Convertibles benefit from some of the risk characteristics of fixed interest securities, having their income and redemption features. They also participate in some of the performance characteristics of shares, due to the conversion option.

The 'vanilla' structure

As with other bonds, a convertible typically pays coupons and carries the promise of cash redemption at maturity. A convertible differs from straight debt in that the holder has the right to exchange the bond, usually at any time, for a predetermined number of the company's ordinary shares, without any extra payment.

On conversion, the investor renounces title to the corporate bond and any future income streams from it in favour of ownership of the predetermined number of shares. A convertible with just this simple structure, with no complications, is sometimes referred to as 'vanilla'.



Vanilla structures are rare, however, as most convertible bonds include call features or other characteristics. Below is an example of a simple callable convertible. The example is used to provide a list of terms useful in explaining and valuing convertibles bonds in general. We use a hypothetical convertible bond, the *Chimera Convertible EUR 4% 2011*. Figure 1 shows the terms of the bond at issue and figure 2 gives some details of the notional underlying share.

Figure 1. Chimera 4% 2011 — Convertible Data at Issue

Coupon	4%
Coupon Frequency	2 (semi-annual)
Issue Date	1 January 2006
First Coupon Date	1 July 2006
Maturity	1 January 2011
Nominal Value	Eur1000
Issue Price	100
Redemption Value	100
Conversion Ratio	10 shares per bond
Conversion Price	Eur100
Call Protection	Callable after 3 years subject to 130% trigger
Call Price	100
Put Feature	-
Issue Size	Eur300 million

Source: Citigroup.

Figure 2. Chimera — Ordinary Share Data

Stock Price	Eur80
Stock Volatility	25%
Dividend per Share	Eur2
Dividend Yield	2.5%

Source: Citigroup.

Figure 3 displays some characteristics of the convertible at issue, based on the data in figures 1 and 2.

Figure 3. Chimera 4% 2011 — Convertible Measures at Issue

Convertible Price	100
Parity	80
Premium	25%
Current Yield	4%
Yield To Maturity	4%

Source: Citigroup.

The meaning of the terms employed in Figures 1, 2 and 3 are explained below.

Terminology

Name convention

Convertible bonds are usually identified by issuer name (Chimera), currency (EUR), coupon (4%) and maturity date 2011.

Coupon

Coupons are the interest payments made on a bond by the issuer, fixed at issue. By convention the coupon is quoted as a percentage of the bond's nominal value. For the *Chimera Convertible EUR 4% 2011*, the coupon is 4% or €40 per bond of €1,000 nominal. As the *Chimera Convertible EUR 4% 2011* pays coupons semi-annually, the coupon will be paid in two instalments of €20 each in July and January.

Accrued interest

Convertible bonds typically observe a 'clean price' quotation convention. (French convertibles are exceptions; see below). When an investor buys a bond at a certain quoted price, he will pay an incremental sum if the settlement date falls between two coupon dates to take account of the accrual of coupon.

French convertible bond prices are normally quoted in 'dirty price' format (inclusive of accrued interest) and 'price per bond' terms, not as a percentage of the nominal value. Thus the bond price will fall by the extent of the coupon, on the date it goes "ex coupon", other things being equal. By convention the "ex coupon" date for French convertibles is the payment date itself, not the last trading day for which settlement falls before the coupon payment date.

Maturity

The maturity date is the date on which the issuer is obliged to redeem unconverted bonds.

The *Chimera Convertible EUR 4% 2011* would mature on 1 January 2011 at 100% of its nominal value or €1,000 per bond. However, it would be rational for holders of the bonds to convert them prior to the final maturity date if the market value of the shares into which they convert exceeds the cash redemption value of the bond.

Nominal value

The nominal (or *face*) value of a bond is its unit size. *Chimera Convertible EUR 4% 2011* convertible bond has a nominal value of €1,000.

Many convertibles in Europe have a nominal value of 1,000 (in the relevant bond currency). Even though there are many exceptions to this rule, 'one bond' is often used to mean '1,000 nominal'¹. As described above, the convertible's price, issue price and redemption value are normally expressed as a percentage of the bond's nominal value; French convertibles are an exception to this rule, being quoted in unit form.

Issue price

When the issue price of the convertible is equal to the bond's nominal value, the convertible is said to have been 'issued at par'. The term 'par' is generally used to mean 100% of the face value of a bond. The *Chimera Convertible EUR 4% 2011* was issued at par, as its issue price (€1,000) was equal to 100% of the convertible's nominal value.

If a bond is issued at a price below its nominal value, it is referred to as an '*original issue discount*' (OID) bond. Likewise, the term 'original issue premium' can be used to describe those issued above par.

Market forces will cause the price of a convertible bond to deviate from its nominal or face value over its life.

¹ There is a lack of consistency in market terminology. Whereas 'fifty bonds' would mean 50,000 nominal (irrespective of the actual nominal per bond stated in the prospectus), one might also hear "one million bonds" taken to mean one million nominal.

Conversion price/conversion ratio

The price at which underlying shares are indirectly purchased, assuming conversion takes place and the convertible has been bought at par, is shown by the '*conversion price*'. The conversion ratio is the number of shares into which each bond carries rights of conversion. For a single currency convertible issued at par² the relationship between the conversion price and the conversion ratio is given by the following formula:

$$\text{Conversion price} = \text{nominal value} / \text{conversion ratio}$$

The conversion ratio for the *Chimera Convertible EUR 4% 2011* is calculated as follows:

$$\text{Conversion price} = \text{nominal value} / \text{conversion ratio}$$

$$\text{Conversion price} = \text{€1,000} / 10$$

$$\text{Conversion price} = \text{€100}$$

There are parallels between the concept of conversion price and the strike price of an option. In a single currency convertible with redemption at par, it will be rational to convert rather than allow the bonds to mature if the share price is over the conversion price³. Thus the convertibles market borrows some terminology from the options market:

- If the share price is over the conversion price, the convertible is said to be *in-the-money*;
- If the share price is under the conversion price, the convertible is said to be *out-of-the-money*.

The share price of Chimera at the time of the convertible issue is €80. As the conversion price on the *Chimera Convertible EUR 4% 2011* (€100) is above the share price of the stock, the convertible is said to be *out-of-the-money*, the normal state of affairs for a convertible at issue.

² When the issue price is below par the conversion price is normally thought of as being a 'moving target'. Initially it is the indirect price per share for one buying at issue price; at maturity it is the indirect price per share for one buying at par. As with dual currency bonds, with original issue discount convertibles it is often clearer to refer to the conversion ratio than the conversion price.

³ In practice a holder who elects to redeem rather than convert may receive an extra coupon, so the decision should take this into account as well. In this circumstance the break-even share price will be (redemption price plus coupon)/conversion ratio.

Parity

Parity is the market value of the shares into which a convertible bond may be converted. Parity is expressed in terms of the bond currency and is normally quoted as a percentage of par. In French bonds it is generally quoted in 'per unit' form.

$$\text{Parity} = \text{conversion ratio} * \text{current share price}^4$$

For the *Chimera Convertible EUR 4% 2011*, parity is calculated as follows:

$$\text{Parity} = 10 * \text{€}80$$

$$\text{Parity} = 800 \text{ (per bond of €1,000 nominal)}$$

$$\text{Parity} = 80\%$$

In a single currency convertible, parity moves with the underlying share price. In a cross-currency bond (where the bond currency differs from that of the underlying equity) movements in the exchange rate between the stock and bond will also influence parity.

Premium

The percentage premium is the difference between the bond's market price and its equity value or parity, expressed in percentage terms.

$$\text{Premium} = \text{Convertible Price} / \text{Parity} - 1 \quad [\text{or } (\text{Convertible Price} - \text{Parity}) / \text{Parity}]$$

The premium on the *Chimera Convertible EUR 4% 2011* is:

$$\text{Premium} = 1000 / 800 - 1$$

$$\text{Premium} = 25\%$$

Premium can also be calculated in per share form. Thus:

$$\text{Premium} = \text{Price of one share bought via the convertible} / \text{Share price} - 1$$

⁴ Expressed in the currency of the convertible

Where:

Price of one share bought via the convertible = convertible price/conversion ratio

In this case:

Price of one share bought via the convertible = 1000/10

Price of one share bought via the convertible = 100

And:

Premium = 100/80 - 1

Premium = 25%

‘Absolute premium’ is the difference between the bond’s price and parity. It is used in some calculations such as ‘breakeven’ (see below)

Current or ‘running’ yield

The current yield of a convertible is similar to the dividend yield of a stock:

Dividend yield = dividend per share / current share price

Running yield = coupon / current convertible price

At issue, the running yield on the *Chimera Convertible EUR 4% 2011* was:

Running yield = coupon / current convertible price

Running yield = €40 / €1,000

Running yield = 4%

A convertible’s current or running yield will change as the convertible price varies.

For a convertible that by convention is quoted with accrued interest included in the price, such as a French bond, analysts will sometimes use a notional clean price to calculate a current yield that is comparable with those of bonds that are quoted 'clean'.

Yield advantage

Convertible yield advantage = convertible running yield – equity dividend yield

For the *Chimera EUR 4% 2009*:

Convertible yield advantage at issue = 4.0% - 2.5%

Convertible yield advantage at issue = 1.5%

Breakeven

'Breakeven' on a convertible may be thought of as the length of time (in years) that it takes for a convertible's extra income to compensate for its premium. 'Breakeven' is a crude measure that makes no adjustment for dividend growth or the present value of future cash flows.

*Breakeven = absolute premium / [coupon – (dividend per share * conversion ratio)]*

For the *Chimera Convertible EUR 4% 2011*:

*Breakeven = €200 / [€40 – (€2 * 10)]*

Breakeven = 200 / 20

Breakeven = 10 years

Since the *Chimera 4% 2011* only has a five-year maturity, and a breakeven of ten years, it is clear that its price cannot be justified on income grounds alone.

Yield to maturity (YTM)

A bond's yield to maturity (YTM) is the rate of return that an investor will receive if the bond is held to maturity⁵.

Bond floor or investment value

A convertible's bond floor, or investment value, is calculated by considering its fixed income attributes in isolation. The investment value of a convertible is found by discounting to present value its coupons and redemption value. This is the same calculation that would be applied to a normal fixed income security to determine its value.

The discount rate generally applied to a bond's cash flows is the sum of the risk-free rate relevant for the maturity of the bond and a credit spread, which reflects the credit quality of the issuer⁶. Without its conversion option, a convertible would be worth no more than its investment value.

Using the risk-free rate plus credit spread method described above, the investment value on the *Chimera Convertible EUR 4% 2011* bond at issue was 89.91% of its par value.

The Libor, or swap curve (for the relevant currency) is conventionally used in the Euroconvertible market as a proxy for the risk-free rate⁷. When choosing a credit spread, one is forced to make an assumption on the credit quality of the issuer, for example by referring to credit derivative or corporate bond markets.

In theory, independent of the performance of the underlying share price, the convertible should be worth at least its bond floor. However a dramatic fall in the share price of a company can damage the issuer's perceived credit quality, which will tend to lower the bond floor.

⁵ For a fuller treatment of *YTM*, please see the Appendix.

⁶ An alternative method for calculating the bond floor considers the spot yield curve and discounts each future payment using the appropriate spot rate plus credit spread.

⁷ Ideally sovereign debt of the nation associated with the currency in question should be used. Markets increasingly adopt the swaps curve as a benchmark, however, for reasons of practicality.

Call protection

A call gives the issuer the right to redeem a convertible before maturity (from the call date) at a predetermined price (the call price). The call price of a bond is often par. In OID or premium redemption securities it is typically at the accreted price.

The *Chimera 4% 2011* can be called at par (100% of its nominal value). However, it cannot be called for at least three years (the bond has three years of ‘*hard call*’ protection). Thereafter there is a ‘*soft call*’ period, during which the convertible can only be called if the share price has reached a specified level. Our notional bond has a two-year ‘*soft call*’ period. The issuer can only call the bond if the stock trades at 130% of the initial conversion price for 20 consecutive days⁸.

When a bond is called, investors almost invariably⁹ have the right to choose whether to accept the call price plus accrued interest¹⁰, or to convert into the underlying shares. In making this investment decision, investors should compare the value of the call price plus accrued interest with the current market value of the shares they would receive on conversion (parity). (Note that on conversion, investors normally forfeit any accrued interest).

If parity is greater than the call price plus accrued interest, the issuer’s decision to call the bond back will, in effect, force investors to convert into shares.

The intention behind the issuance of a call notice is frequently to force conversion. For this reason the issuer may delay calling the bond until a ‘cushion’ has developed through parity rising to a level materially in excess of the call price. In this way the possibility of share price weakness during the notice period making conversion sub-optimal is reduced.

⁸ Note that the actual call price is of little importance in soft call bonds if it is well under the trigger level. Since the bond is worth 130 (say) converted, by definition, it does not really matter if the call price is 100 or 105 for example, as rational investors will convert.

⁹ There have been rare instances of a bond being callable before it can be converted.

¹⁰ Assuming the call notice specifies payment of accrued interest. In instances where the call date falls on a coupon payment date, this issue does not arise.

Sometimes it may be rational for an issuer to call a bond even if parity is well below the call price. This may be the case either if interest rates have fallen or if the company's credit has improved, placing the company in a position to refinance on more attractive terms. Calls for cash redemption are less common than calls forcing conversion.

Put features

While most convertibles have call features, investor put options are less common in Europe. A put feature gives the holder of the convertible the right to require the issuer to redeem a convertible on a predetermined date prior to maturity, at a fixed price. The issuer is usually required to redeem the convertible for cash. However, some convertibles give the issuer the option of delivering shares, or a mix of cash and shares. When settlement is in shares the term 'soft put' is sometimes employed.

Convertibles as derivatives

Equity investor viewpoint

A convenient decomposition of the payoff of a convertible bond for the equity investor is:

- **Equity** — a convertible holder has control over a fixed number of shares. This equity value is represented by parity.
- **A put option** — a convertible holder has control over the underlying shares, but if the stock price falls, the investor can choose not to receive the shares but a cash redemption amount at maturity. The right *not* to convert is equivalent to a put option; imagine the investor automatically being delivered the fixed number of shares at maturity and then having the right to sell the package of shares back at the cash redemption amount.
- **A dividend swap** — a convertible holder indirectly owns a certain number of shares, but receives coupons instead of dividends until the earlier of maturity or conversion. This is equivalent to owning a single name dividend swap that gives the equity investor fixed payments (the coupons) in exchange for dividends. Assuming there is an income advantage for the convertible holder, the investor will pay a premium to own this embedded dividend swap.

Bond investor viewpoint

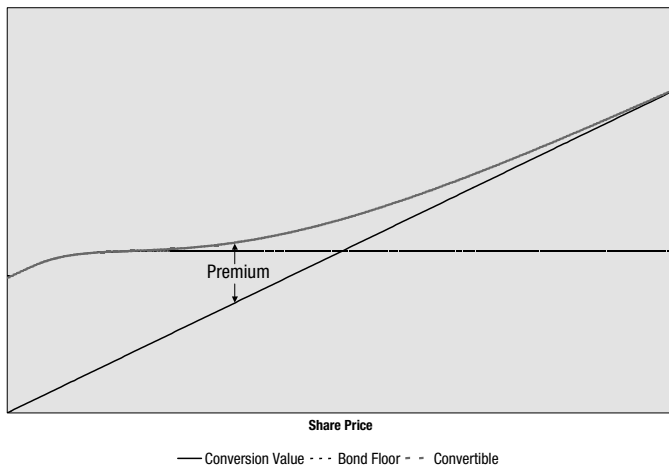
Fixed income investors tend to view convertibles as a combination of:

- **A straight bond:** a convertible holder owns a straight corporate bond that pays coupons and has a cash redemption value at maturity. A convertible's bond floor represents this straight bond value.

- **A call option:** a convertible holder has the right to exchange his corporate bond for a predetermined number of the company's ordinary shares, usually at any time. This is a call option that gives the fixed-income investor the right to acquire shares by giving up the convertible. At maturity it is equivalent to a call with a strike price equal to the conversion price (for a single currency par redemption bond). In contrast to the exercise of a conventional option, however, there is no cash payment on conversion.

Figure 4 displays how the value of a convertible might vary with the share price some time before maturity.

Figure 4. Premium Against Share Price



Source: Citigroup.

It is of note that:

- The bond floor of a convertible is not correlated with movements in the share price, unless a fall in the stock price is perceived to influence the credit quality of the issuer (shown in the extreme left of Figure 4).
- Parity or 'conversion value' varies proportionally with the share price, being the product of the share price and the conversion ratio.
- Figure 4 shows how investors purchasing shares via a convertible forfeit a degree of equity appreciation, the convertible price having less than 100% share sensitivity through most of the spectrum.
- The difference between the convertible price and parity (premium) represents the price of the embedded put option (downside protection) and the dividend swap (yield advantage) to an equity investor.
- Figure 4 illustrates how, as the equity price rises, the premium narrows to a point where the convertible derives its value exclusively from the equity component (apart from income factors).
- When the equity price declines, the premium expands. The more the share price falls, the more the convertible price is supported by and trades in line with its bond floor.

Valuation stimuli

A convertible's theoretical value is positively correlated with:

- *The underlying stock price* — the higher the underlying stock price, the greater the value of parity (the equity component of a convertible), so the greater the value of the convertible.
- *Volatility of the underlying stock* — the higher the volatility of the underlying share, the greater the value of the convertible's embedded option (put or call, according to the equity or the fixed-income investor's perspective), so the greater the value of the convertible.

- *Issuer credit quality* —the better the credit quality of the issuer, the lower the discount factor applied to a bond's cash flows, the higher its bond floor and the greater its value.
- *Call protection* — the longer the call protection on a bond, the longer the period of any yield advantage in a convertible. Longer options also tend to be more valuable than shorter ones.

A convertible's value is negatively correlated with:

- *Stock dividends* — the higher the dividend, the lower the yield advantage of the convertible over the share.
- *Interest rates* — the higher the prevailing interest rate, the higher the discount rate for coupon payments and redemption value, the lower the bond floor and convertible's value. It is also true that the higher the interest rate, the greater the value of the convertible's embedded call and so the greater the convertible's value. Thus a convertible's price tends to be less sensitive to interest rates movements than an otherwise identical straight-bond.

Greek or Greek sounding names are conventionally applied to the impact of small changes in various determining variables on the value of a convertible.

Delta

Delta describes the sensitivity of a convertible's value to changes in parity. It is a measure of the equity sensitivity of the convertible.

Conventionally, delta is expressed as the change in convertible price for a unit change in parity. Thus, a 40% delta means that if parity rises by one point, the convertible value will rise by 0.4 points (40% of the one-point change in parity)¹¹.

¹¹ To be precise it is the sensitivity to a very small absolute move in parity rather than a one point move (see appendix).

It is important to note that delta does not deal with percentage movements. If the convertible responds to a 1%

Gamma

Gamma is the rate of change in delta for movements in the underlying share price. It is often expressed as the change in delta for a one-point change in parity. There is no uniformity of convention for gamma however. Some use the term to refer to the sensitivity of the delta to a 1% change in the share price, for example.

For significant share price moves, delta can be a poor guide to the sensitivity of the convertible. In general the convertible is more equity sensitive in rising markets and less sensitive in falling markets. Gamma is the measure of the intensity of this effect. This topic will be covered later when discussing '*gamma trading*'. A theoretical treatment of gamma and a chart of its sensitivity to the share price level are available in the Appendix.

Vega

Vega is the sensitivity of the convertible value to changes in the volatility of the underlying stock. Conventionally, vega is expressed as the change in the fair value of the convertible for a one percentage point increase in the assumption for stock volatility.

In order to find a theoretical value for a convertible, it is necessary to enter a number of assumptions into a convertible pricing model:

- A credit spread (spread over Libor or Treasuries), used to establish the rate for discounting the bond's coupon payments and its redemption value;
- Dividends on the underlying share, to maturity;
- Stock borrow costs and
- A volatility assumption for the underlying stock.

parity move with a 0.5% move this is NOT a 50% delta (unless the premium is zero), eg parity goes from 80 to 80.8 (1%) and CB goes from 100 to 100.5 (0.5%), delta is 62.5% (0.5/0.8).

Vega measures the sensitivity of fair value to changes in the assumed level of stock volatility.

In approximate terms the closer the convertible is to being at-the-money, the more sensitive its value is to changes in assumed volatility. A chart of vega against parity is available in the Appendix.

There are no short names in common use for the dividend sensitivity or stock borrow cost sensitivity. Spread sensitivity is sometimes referred to as CR01.

Rho

Rho¹² measures the sensitivity of convertible value to movements in interest rates. Conventionally, it is expressed as the change in convertible price for a one basis point move in interest rates (a parallel shift in the whole yield curve).

Rho is the fixed income sensitivity of the convertible just as delta was defined as the equity sensitivity.

Delta increases as parity (the equity component of the convertible) increases. Conversely, rho tends to increase when parity decreases. As the share price falls, the convertible derives an increasing proportion of its value from its fixed interest attributes.

A more thorough treatment of Rho is to be found in the Appendix.

¹² Also known as DV01.

Theta

Theta is the change in convertible value due to the passage of time. It is not identical to the term as used in the options market as it combines the redemption pull of a bond with the time decay of an option.

Conventionally, it is expressed as the percentage change in convertible price for the passage of one day, other things being equal. For a near-the-money convertible, the passage of time is normally negative for its value, the time decay of the option element outweighing any upward drift in the bond floor.

Convertible Bonds: Equity Investor Perspective

- Risk considerations
- Switching shares into convertible
- Preferred habitat

Risk considerations

Equity investors will be attracted by the asymmetrical equity sensitivity of convertibles. As Figure 4 shows, there is a tendency for the convertible to become more equity sensitive in a rising share price environment and less so when the share price is falling. This gives the asset class attractive risk characteristics that are not always captured by conventional measures. Measures such as standard deviation, which identify risk with the incidence of extreme observations in *either* direction, implicitly assume that the overall return profile is symmetrical. To take an extreme example of the power of asymmetry, consider a notional instrument that can go up but not down. The standard deviation of expected returns may be high, but the instrument is not really 'risky', as most would understand the term.

Switching shares into convertible

When an equity investor wishes to sell shares and replace them with convertible bonds, there are three common methods available for selecting the appropriate quantities to deal in

- Retaining control of same number of shares
- Cash-for-cash
- Applying a delta

Retaining control of the same number of shares

This is a particularly interesting strategy for those who are uncomfortable with the loss of equity exposure that is typically involved in switching from shares to convertible. In effect it amounts to applying additional funds to the extent of the premium. In cases where the convertible looks cheap on income grounds alone there is particularly good justification for this approach. The move can be likened to an indirect repurchase of as many shares as are being sold and simultaneous investment in incremental income (much like buying an annuity) for the period up to expected conversion.

Cash-for-cash

The cash-for-cash switch is a less common strategy for equity investors than used to be the case. Paying a premium for the convertible involves the sacrifice of some involvement in the share.

Static delta

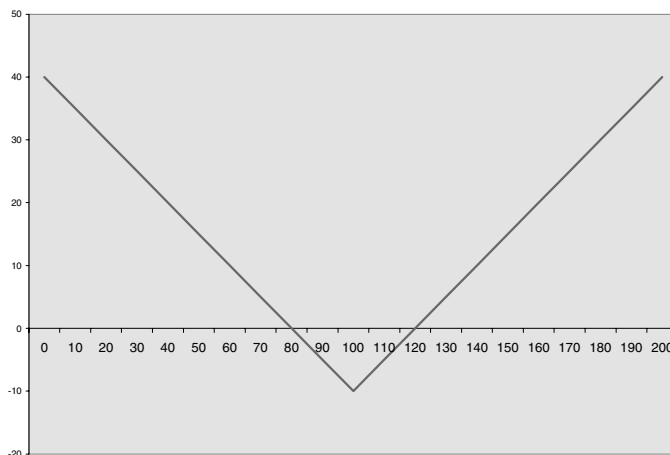
By mimicking the behaviour of the hedge fund and attempting to maintain equity sensitivity, the equity investor will typically have to put considerable additional funds to work. Unlike the hedge fund, the equity investor will not normally adjust the ratio over time. This means he can estimate at the time of the switch the circumstances under which the trade will prove to be profitable.

Take a convertible bought at 100 on which parity is 80 and the delta 50%. For every €100 invested in the convertible the sale of shares will only raise €40 (since selling share on a 100% delta would raise €80), so the balance of €60 must be borrowed or taken out of cash balances.

The investor can now employ breakeven arithmetic based on the assumption that the convertible will either end up at parity or at the redemption price. There is a zone of loss in the event that the share price shows little movement but increasingly significant gains on extreme share price movements.

The payoff of the whole position (long convertible, short cash, short shares on a delta) is shown in Figure 5.

Figure 5. Payoff Profile of Convertible Hedged on a Static Delta



Source: Citigroup.

This type of analysis can provide useful insights into the relationship between share and convertible even for those who will in practice be adjusting the delta. The approach involves considering all characteristics of the portfolio consisting of ‘long the convertible and short a given number of shares’, including the net cost of financing the position after stock borrow costs, coupons and dividends have been taken into account.

Preferred habitat — in-the-money convertibles

Convertibles on very low premiums are often candidates for equity investors, as they are a better proxy for the shares themselves. Such convertibles typically have high deltas and a relatively remote investment value.

Some equity investors are attracted by the income characteristics of certain convertibles. This is particularly so if the premium paid is made up for by an income advantage over the underlying shares. Instances in which the premium is recovered in incremental income before the first call date of the convertible have evident appeal (see the definition of ‘*breakeven*’).

Convertible Bonds: Bond Investor Perspective

- **Premium to bond floor**
- **Expected yield to maturity**
- **Preferred habitat**

Premium to bond floor

For a fixed income investor, the purchase of a convertible is equivalent to the purchase of a straight bond together with the purchase of upside participation in any underlying share price appreciation. The investor pays the premium to the bond floor in order to gain access to this participation.

When the underlying share price rises, the premium to bond floor of the convertible will tend to expand due to the equity participation element. Similarly when the underlying share price falls, the bond floor premium tends to contract. The convertible price will fall, but as long as the fall in the underlying share price does not affect the credit quality of the issuer, the convertible's bond floor should support its price.

Bond floor premium expansion can be interpreted as the potential advantage that a fixed income investor stands to gain by switching from a straight bond into a convertible.

Conversely the premium to bond floor describes the amount that a fixed income investor stands to lose from switching a straight bond into a convertible, if the underlying share price falls. Looked at in this way there are parallels between the role of premium to parity in the psychology of the equity investor and premium to bond floor in the psychology of the fixed interest investor.

Expected yield to maturity

Conventionally the yield to redemption of a convertible is calculated using the formal redemption price of the instrument. However a concept of ‘expected yield to maturity’ is a useful one when looking at convertibles from the standpoint of the fixed interest investor. In this case one ascribes probabilities to different share price outcomes and thus different effective bond redemption prices. One arrives at an ‘expected redemption price’ and calculates an expected yield to maturity accordingly. This is very similar to the classical options approach to convertibles valuation except that in this case one does not solve for fair value, one solves for the expected internal rate of return given the current price of the convertible.

Preferred habitat — out-of-the-money convertibles

Traditionally, fixed-income investors with a low risk profile have preferred to buy out-of-the-money convertibles since they tend to trade close to their bond floor. Such bonds respond to movements in interest rates and credit spreads in a manner similar to that of straight bonds.

In some cases, fixed income investors may not be motivated at all by upside participation in underlying share appreciation and may only be interested in the bond component of the convertible. In this case they will buy deep out-of-the-money convertibles, as they will be reluctant to pay any premium to the perceived bond floor. This type of investor may have detected a yield advantage in convertibles compared to equivalent straight-bonds due to anomalies in the market. The relative coupon structure and sometimes the subordination of the convertible can differentiate it from other debt of the issuer.

Convertible Bonds: Hedge Fund Perspective

- **Isolating determinants**
- **Hedging equity risk**
- **Hedging currency risk**
- **Hedging yield curve exposure**
- **Hedging vega risk**
- **Hedging dividend risk**
- **Hedging credit risk**

Isolating determinants

As discussed in the previous section, the value of a convertible bond is sensitive to the behaviour of a number of variables including:

- Share price
- FX (for cross currency bonds)
- Yield curve movements
- Option valuations
- Dividends
- Credit spreads
- Stock borrow costs

One way of describing the difference in the attitude of the traditional unhedged investor and the hedge fund is that the former makes investment decisions based on his view on the combined impact of all these variables, whereas the hedge fund will choose to take a view on a fewer number of variables by neutralising his sensitivity to the others.

At the extreme, in a ‘pure arbitrage’ situation, the hedge fund will seek to sterilise all influences on a cheap convertible to lock in a risk-free profit.

Hedging equity risk

It is possible to construct a portfolio that is theoretically insulated from small share price movements. This portfolio is based on a long convertible position and a short position in the underlying share. The number of shares that must be sold is indicated by the convertible’s delta or ‘*hedge ratio*’.

Delta hedging and gamma trading

As the share price underlying a convertible rises, the delta will also rise, meaning that more shares need to be sold to keep the portfolio delta neutral. As the underlying share price falls, the convertible’s delta falls and some shares must be bought back in order to re-hedge the position. Thus hedged investors tend to buy shares after a price fall and sell after a price rise. This process of rebalancing, termed ‘gamma trading’ is one way in which hedge investors seek to generate profits from their portfolios.

Figure 6 uses the details of the notional *Chimera EUR 4% 2011* issue, used above.

Figure 6. Chimera Convertible 4% 2011 Long Convertible Position

Convertible Price	Eur1000 (100%)
Conversion Ratio	10
Underlying Share Price	Eur80
Delta	40%
Gamma	0.5%

Source: Citigroup Convertibles.

In order to hedge the long position of one bond against changes in the underlying share price, one needs to take a short position in 4 shares:

Short share sale = conversion ratio * delta

Short share sale = 10 * 40%

Short share sale = 4 shares

The hedge fund will take into account the following:

- The greater the move in the underlying share price, the greater the chance of significant gamma¹³ trading profits.
- Stock volatility is a measure of the propensity of the underlying share to move sharply. The higher the underlying stock volatility, the greater the potential profits from the hedged convertible position.
- ‘Implied volatility’ is the stock volatility which renders the convertible fair value at its current market price. Those expecting the share to exhibit greater volatility than this will find the dynamic delta hedging strategy appealing.
- When the share fails to exhibit the required level of volatility the resultant failure of the strategy comes as a result of time decay of the embedded option.
- The choice of hedge ratio is by no means simple. If the hedged investor believes the convertible to be mis-priced, he must decide whether to hedge on the model output given his own volatility assumption or on model output using the implied volatility¹⁴. He may also look at the ‘empirical volatility’; the historical relationship between parity and share price suggested by regression analysis.

¹³ Recall gamma measures the sensitivity of delta to movements in the underlying share. The term ‘gamma trading profit’ describes the gains to be made from dynamic delta rehedging.

¹⁴ For an extreme example, consider a convertible on a zero implied volatility. Does the investor want to take the free option and not hedge it or hedge on the basis of what he thinks the option is worth? Probably the latter in our view, which argues in favour of hedging on the basis of ones own assumed volatility.

Hedging currency risk

The hedge fund's easiest way of dealing with currency risk in a cross currency bond is to put on the equity hedge by selling shares denominated in the currency of the bond. This leaves the investor with the currency risk of a single currency bond (i.e. none at all if he is measured in terms of the bond currency). Where no such ADR type instrument is available, it is sometimes possible to deal in one synthetically through a cross-currency equity swap.

For those (including the issuer) who wish to know what the FX exposure is in isolation, the problem is complex. At the extremes the situation is reasonably clear. A 100% delta convertible is pure equity; to hedge one sells all the shares into which one has conversion rights and carries out any spot foreign exchange transaction needed to receive the proceeds in the investor's base currency¹⁵. If the convertible is on a zero delta, the instrument is a pure fixed interest instrument. If the bond is denominated in a currency other than the investor's base currency he can in theory hedge the exposure by executing forward transactions on all future payments from the bond.

Between these extremes it is necessary to estimate probabilities for different foreign exchange requirements at different dates and hedge these accordingly. This is naturally not a static hedge. In order to generate the probabilities a Monte Carlo simulation approach can be employed¹⁶.

¹⁵ There may remain some residual currency risk relating to the coupons; these would have to be dealt with using one of the techniques discussed.

¹⁶ See *Analysing Currency Risk in Cross-Border Equity-Linked Bonds*, Bapi Maitra and Eu Jin Ang, of CitFX, RISK Magazine March 2002.

Hedging yield curve exposure

Consider a vanilla non-callable convertible¹⁷ in a flat yield curve environment. The pay-off can be replicated as a corporate bond and a call option. Each of these has its characteristic yield sensitivity, the bond benefiting from yield declines and the option component benefiting from yield increases¹⁸. The bond component will prove the prevalent force, leaving convertibles with yield exposure of the same direction as straight bonds, but of a lower magnitude.

Complications arise from the fact that yield curves are not flat and from the lack of certainty over which points in the yield curve are of relevance. In a coupon-paying bond there is some yield exposure at every coupon date until conversion or redemption as well as the exposure at that date itself. If the expected life of a convertible falls short of the final maturity date, the latter may not be an important yield curve point. Many practitioners obtain model output for yield exposure in given 'time buckets' for individual positions and aggregate across a portfolio.

Hedging vega risk

Although in many cases a hedge fund will retain the option element of a convertible and hedge this dynamically against the share, on occasions it will wish to hedge its option value exposure. This may be the case, for instance, if it wishes to lock in the profit from a pricing anomaly between pure options and those embedded in a convertible.

It is normally impossible to replicate the exact option embedded in the convertible. This is because of possibility of early exercise and the fact that the strike price, being the value of the bond that is given up on exercise, is not a set quantity. Sometimes approximations can be made using the fugit (or expected life). Even here there are technical and practical difficulties.

¹⁷ Further assume there are no rights of early conversion so that it is a European option.

¹⁸ Other things being equal, the higher the interest rate the higher the forward price and the more a call option is worth.

One theoretical problem is that the fugit life date can be very imprecise. For example, if the convertible has a 50% chance of lasting one year and a 50% chance of lasting five years the fugit life will be three years, but hedging with a three-year option would not be very accurate¹⁹. A further theoretical problem lies in the difficulty of attaching a number to the appropriate strike price for the equivalent option. If the adopted horizon is the maturity date, the strike in per share terms will be the redemption price divided by the conversion ratio²⁰, but if it is some other date it will be the then prevailing bond floor divided by the conversion ratio. This bond floor can only be estimated.

Even if the investor is satisfied as to the specification of the nearest option to that embedded in the convertible, there remains the practical problem that no options of that description may be available. For example, one may conclude that a four-year 100% out-of-the-money call is the appropriate contract, but discover that strikes more than 50% away from the money or with more than two years to final expiration are not available in the share in question. In this case one approach is to match the vega of the convertible and an option of a different specification. This would leave the investor with new exposures to interest rates, share prices and time decay to deal with.

When comparing pure options with convertibles the investor should be conscious of any adjustments that need to be applied to allow for any convention differences between the two markets. The convention in options markets is to state implied volatilities in a risk free environment and without stock borrow costs, whereas many convertible models employ risky rates and take account of stock borrow costs. It is good practice to use the price not the quoted implied volatility of a given option and work this back to an implied volatility using similar conventions to those employed in the user's convertible model.

¹⁹ One way of checking the precision of the fugit number is to check it against 'reciprocal fugit' or the harmonic mean life. The closer the two measures, the more precise the fugit estimate.

²⁰ Except in the case of single currency bonds with redemption at par this is not necessarily the conversion price. Even in this case an adjustment to the redemption price to account for the final coupon may be called for.

Hedging convertible dividend risk

If the convertible is looked on as a bond plus an option, all the dividend risk lies in the option component. If the vega is hedged through the sale of a similarly specified pure option, the dividend risk will also be eliminated²¹. The representation of the convertible as a share plus a put plus an income difference makes overt reference to the dividend component allowing it to be hedged explicitly.

The natural mechanism to hedge this exposure is the dividend swap. An investor can contract to pay fixed amounts for a fixed period in exchange for variable dividend related amounts. The major practical difficulty is that an investor seldom knows for certain how many dividend payments he wishes to swap. He must either accept a possible mismatch or enter into a more complex cancellable contract.

²¹ Though obviously differences in the definition and treatment of special dividends would need to be watched.

Hedging credit risk

Hedging credit risk — callable asset swaps

A callable asset swap seeks to repackage the convertible such that one party owns only the fixed income component of the security and the other owns the option component.

In reality a callable asset swap is not a swap at all, as there is no exchange of future flows. It is a contract in which the option investor sells the convertible to the bond investor below the market price and retains the right to buy it back. The bond investor acquires an asset with most of the interest rate and credit characteristics of a normal bond. The option investor, through his call option on the convertible, mimics the condition of having a call on the underlying shares. The objective is for the option buyer to have no exposure to the credit of the underlying share and no interest rate exposure except that of a stock warrant.

The terms of the contract specify the cash amount due for selling the convertible and the formula for calculating the price at which the convertible may be bought back. The cash amount corresponds to the value of the convertible's bond floor, calculated at a given credit spread over Libor²². The price at which the convertible may be repurchased is worked out as another bond floor, calculated using a lower credit spread.

The credit spread at which a fixed income investor buys the bond element of a convertible in a callable asset swap may be higher than the credit spread implied in the price of a straight-bond with the same profile. Thus, a fixed-income investor may be able to buy a straight bond proxy at an advantageous yield.

To understand why it is a reasonable proxy in most circumstances, one must look at the situation of the bond investor at the maturity of the bond. The option investor has the right to buy the convertible at (about) the redemption price. Whether or not this right is exercised, the bond investor is assured of getting the redemption price at maturity (default excepted).

²² The price is typically calculated by reference to asset swap market methodology.

If parity is under 100 he will expect to get it from the maturity of the bond and if parity is over 100 he will expect to get it from the swap counterparty.

There are ways in which the callable asset swap fails in its objective of delivering a bond equivalent to one party and an option equivalent to the other, however. Pronounced spread narrowing may mean that the bond investor may be put in a situation in which he is required to sell the bond below the fair market price of a pure bond with similar terms; there is in essence a credit option against him.

Callable asset swaps can also miss their objectives due to complications arising from early recall. This generally involves a cost for the option investors and 'windfall' for the bond investor. To the extent that the option investor pays more than the fair bond floor when it comes to buying the convertible back, he is suffering a penalty. For the bond investor, the possibility of an early call is positive on balance.

Valuing existing callable asset swap contracts

The embedded credit options and early exercise features present interesting valuation problems for the callable asset swap. One mark-to-market technique for the option investor is the intrinsic value approach; the position is valued as if the bond were recalled immediately. At first glance the approach appears to have merits; it is easy to calculate and tends to value the position conservatively. However, it is sometimes so conservative as to be a patent misrepresentation of the true value. For example, it calls for the position to be marked at zero if the bond floor at the recall spread is higher than the market price of the convertible, yet the right to buy the convertible in the future has clear value (which could be realised on reassignment of the contract).

The other end of the conservatism spectrum is to assume that the convertible can be recalled without penalty, that is to say at the actual bond floor. In situations in which the option to recall early is unlikely to be exercised, this is a fair reflection of the value of the contract.

A compromise solution is to include a proportion of the recall penalty, having regard to the probable timing of any early exercise.

Hedging credit risk — credit default swaps

Credit default swaps (CDS) are now the most common medium for transferal of credit risk. Typically, one party in the swap faces credit risk from a certain entity, and the counterparty in the credit default swap agrees to assume this risk in exchange for regular periodic payments. If the reference entity defaults (or on certain other credit events), the party providing insurance will typically have to purchase from the insured party one of a specified basket of assets at face value.

The regular stream of fees paid to the protection seller is typically expressed in basis points per annum ('default swap spread'). The fee ends when the contract is terminated or reaches its final maturity, often five years.

For example, suppose two parties enter into a five-year credit default swap. Assume that the notional principal is US\$100 million and the buyer agrees to pay 90 basis points annually for protection. If the reference entity does not default, the buyer receives no payoff and pays US\$900,000 per year (normally in quarterly installments) for each of the five years.

The convention for quoting CDS prices is as follows: the bid is the price that the market maker would pay to buy protection (sell the risk), and the offer is the fixed rate that he would require to provide protection (take the risk) in physically settled contracts.

Hedging credit risk — buying put options

The idea behind buying out-of-the-money put options to hedge credit exposure is that a company getting into credit difficulties is likely to experience share price weakness. The increasing value of the put options should compensate for the decreasing value of the bond position. Ultimately the payoff from the puts can act as a substitute for the redemption value of the bonds.

For example, an investor owns US\$1 million nominal of a one-year bond. The share price of the issuer stands at 100. One-year puts with a strike price of 25 can be bought for US\$1. The investor buys 40,000 such puts. If the company defaults and the share price goes to zero, the payoff on the put options will be $US\$25 \times 40,000$ or US\$1,000,000, replacing the redemption proceeds on the bond even if the recovery rate is zero. For US\$40,000 the investor has 'insured' his bond position.

Notional put purchase — adjusting the delta

A development of the idea of buying out-of-the-money puts in the market is to calculate the selling activity in the share that the counterparty to such a transaction would undertake. Instead of buying the puts the investor simply sells shares on the calculated delta. In a sense he has sold the put option to himself and hedged accordingly. There is an interesting coming together of older and newer hedging approaches in this process.

Increasing the delta on a convertible position is a conventional expedient for dealing with situations in which a weak share price is likely to be associated with a widening credit spread.

A note on capital structure arbitrage

When assessing the relative attractions of buying credit protection in the credit derivatives market and buying put options, hedge investors sometimes found marked discrepancies. Not surprisingly, a new activity grew up in which protection was simultaneously bought through one medium and sold through another.

In the early days of capital structure arbitrage, the major focus was on the arbitrage limits of certain combinations. For example if it is assumed that default inevitably means a share price going to zero one can easily calculate how many put options would be required to replace the maturity amount of the bonds (see above). Even ignoring any other value coming from ownership of the puts the cost may be much less than the cost of buying credit protection in the credit derivatives market.

The market is now alert to these pure arbitrage conditions and the business of capital structure arbitrage tends to focus on judging whether the implied probability of default in different products is markedly different.

The most contentious matter in computing the implied probability of default from CDS levels regards recovery levels. If zero recovery is assumed, the CDS level itself is an approximate measure of the implied probability of default.

The most common technique for estimating implied default probabilities from option prices works through a line of reasoning that has become known as Merton analysis. The value of the whole firm is seen as the underlying variable and default is identified with the value of the firm falling below the value of the debt. Using volatility estimates for the enterprise derived from those of the share price behaviour, and debt level figures taken from the balance sheet, a probability of default for a given time horizon is estimated.

Convertible Bonds: Issuer Perspective

- **Convertible financing**
- **Meeting objectives**
- **Convertibles and IFRS**
- **To call or not to call?**

Convertible financing

There are mirror images in the view of convertibles from the issuer's point of view and from that of the investor. Per dollar invested investors would normally achieve superior returns by investing directly in shares when the underlying share price rises and would do better owning straight debt when the underlying share price falls. From the issuer's standpoint and with the benefit of hindsight, equity issuance would typically have been superior to convertible issuance if the share price has fallen since issue and bond issuance would have been superior to convertible issuance if the share has been strong.

Limitations

The limitations to this idealised way of looking at the relative merits of bond, equity or convertible are the same for the issuer and the investor: practical considerations and uncertainty.

In the case of the investor one practical matter is likely to be portfolio constraints that limit the proportion of equity or fixed interest paper he may hold. In the case of the issuer, balance sheet structure is likely to be a constraint. Even if the company holds an optimistic view of its own share price, for example, there is a limit to the extent of balance sheet gearing that will be deemed appropriate. The way in which equity linked securities can be employed to solve the practical problems of issuers are discussed below (see *Financing objectives and prospective issuers*).

Uncertainty was the consideration that permitted the investor to consider equity/bond hybrids superior in terms of the overall risk and return profile to either asset class in isolation²³. Likewise the issuer who issues a convertible avoids the ‘worst-case scenario’ of selling debt or equity at a time that will prove to have been unfortunate with the benefit of hindsight.

²³ For the investor, there are parallels with the way that uncertainty governs the theory of portfolio diversification. A fund manager would not normally devote his entire portfolio to his favourite share, no matter how strong his conviction. Likewise he should consider a lower risk lower reward version of a given share however much he favours that share.

Financing objectives and prospective issuers

Alternative to equity funding

The idealised concept of choosing whether to issue equity or debt based on the perceived outlook for the share price is seldom realistic. However, if the choice is limited to convertible or pure equity, share price expectations may well be an important consideration. If the decision takers in an issuing company are confident of share price strength, the issue of a convertible looks very much like the issue of shares at a premium to market levels at the time of issue. The convertible issue carries none of the stigma sometimes associated with a rights issue and is often less cumbersome to execute.

Monetising equity investments

‘Exchangeable’ structures²⁴ can permit issuers to monetise stakes they hold in other companies in a convenient way. The issuer retains voting privileges, and can engineer prospectus terms to give delivery flexibility in the event of conversion or redemption.

Monetising volatility

A company that considers the implied volatility of options on its shares to be unreasonably high can exploit this view by selling options embedded in convertibles. The company’s own share price volatility is an asset that can be bought or sold.

Securing low-cost funding

One attraction of convertibles relative to straight bonds from the issuer’s perspective is the lower coupon that will be commanded. Companies anxious to limit debt-servicing costs may consider the grant of equity optionality an acceptable price to pay.

²⁴ For more information on ‘Exchangeables’, see page 56. Exchangeables are bonds that can be converted into the shares of a company other than the issuer.

Tax/cash-flow advantages

Under most tax jurisdictions, only interest payments on debt are tax-deductible; dividends are paid out of after-tax income. Thus convertible debt financing is often more tax efficient than equity issuance. In the case of zero coupon convertibles, no cash interest is actually paid, but in most tax regimes the company can deduct the interest implied by the accretion rate of the bond, creating a cash-flow advantage. In the case of exchangeable bonds, issuers are allowed to defer the crystallisation of tax liabilities on the sale of shares under many tax regimes, as beneficial ownership of the underlying shares transfers to the investors only on conversion.

Delaying dilution

Compared with the immediate issuance of equity, the issuance of a convertible may be beneficial for earnings per share. This is particularly true in cases where the probability of conversion is low, as analysts will typically employ undiluted EPS calculations. Moreover convertible investors do not have voting rights, so the issuance of a convertible defers the dilution of voting rights in the company. Closely held companies, for whom the issuance of voting shares may compromise control, may have a particular interest in raising quasi equity capital via convertibles.

Exploiting market conditions

One advantage of issuing a convertible is that distribution can frequently be completed in one day, whereas an equity issue typically takes several weeks. Speed of execution is a key advantage of convertible bond issuance.

Targeting a diverse investor base

The geographical and investment-style diversity of the convertible investor base enable the issuer to access a wide investment audience and enhance corporate recognition.

A company that has recently completed an equity issue may find it easier to raise further capital via a convertible rather than returning to the equity market. The size of a convertible issue that the market is capable of absorbing is often impressive.

Convertibles and IFRS

The International Financial Reporting Standards (IFRS) treats the equity derivative component of some European convertibles as “equity” and others as “liability”. Issuers of bonds in the latter category are required ‘mark to market’ the value of the embedded option for income statement purposes.

“Equity treatment” bonds

Included: bonds carrying rights of conversion into the shares of the issuer on fixed terms with no issuer option to deliver cash in lieu of shares on conversion.

Identification of bond portion: The value of otherwise identical bond with no conversion rights.

Treatment of bond portion: The bond portion is treated as any other fixed interest security issued by the company. It is recorded in the balance sheet as a liability. Under IFRS it is revalued according to market conditions at each balance sheet date with adjustments posted to retained earnings. On conversion the ‘liability’ is reclassified as ‘equity’. If there is no conversion it continues to be treated as a bond.

Identification of equity portion at issue: The excess of the proceeds of the issue over the bond portion.

Initial treatment of equity portion: The equity portion is recorded as equity in the balance sheet and remains such irrespective of whether rights of conversion are exercised.

“Liability treatment” Bonds

Included: exchangeables, bonds with issuer’s cash delivery at conversion option, bonds denominated in a currency other than the issuer’s functional currency²⁵.

Identification of bond portion: The value of an otherwise identical bond with no conversion rights (same as with “equity treatment” bonds).

Treatment of bond portion: As with “equity treatment” bonds, the bond element is marked to market at balance sheet date and is reclassified from liability to equity on conversion.

²⁵ In a meeting in September 2005 the International Financial Reporting Standards Board considered and rejected a proposal to amend this ‘foreign currency’ feature of IAS32.

Identification and treatment of derivative portion: At issue the derivative portion is the excess of the proceeds of the issue over the bond portion, thereafter it is marked to market at balance sheet date according to market conditions.

Call decision framework

Often the ultimate goal of a convertible issuer is for its convertible bond to be converted into shares. Call features allow the issuer to precipitate conversion through the dispatch of a call notice. The company in this predicament has a dilemma when deciding whether or not to issue a call notice, however. If parity falls under the redemption value between the date investors are notified of the call and the call date itself, conversion is unlikely to be forced and the company would be required to redeem the security in cash.

For this reason many issuers will require a 'cushion' for the excess of parity over call price, before issuing a call notice.

Throughout the call period, that is the period between the issue of the call notice and last date on which investors must make the conversion election, the investor owns an option against the issuer. This is most easily thought of as being a put option: the investor 'owns' a parity number of shares but has the right to sell them back at the call price.

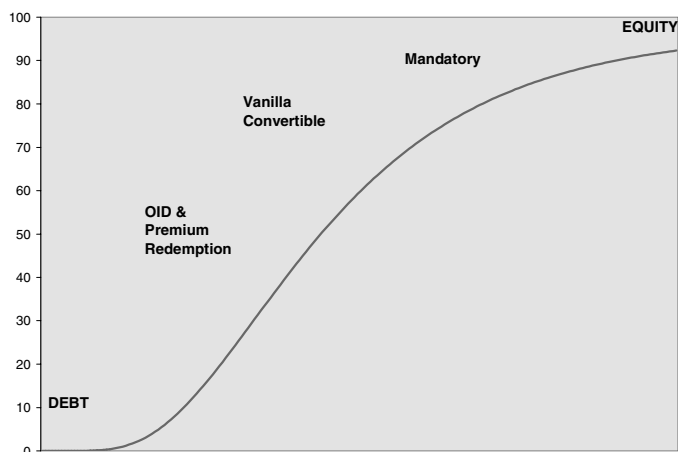
Convertible Structures

- **The debt/equity spectrum**
- **Original issue discount (OID) convertibles**
- **Vanilla convertibles compared with a bond + warrant**
- **Mandatory convertibles**
- **DECs**
- **Exchangeables**
- **Reset features**
- **FRNs**

The debt/equity spectrum

Convertible securities can be structured such that they bear a greater resemblance to debt or equity investments. Figure 7 shows approximately where on the debt/equity spectrum the more common structures lie at issue.

Figure 7. The Convertible Structure Spectrum



Source: Citigroup Convertibles.

Original issue discount (OID) and premium redemption convertibles

Original issue discount convertibles and premium redemption convertibles are near identical structures. In the former the issue price is below par and it is redeemed at par, in the latter the issue price is par and it is redeemed above par. In both cases the probability of conversion is lower than a conventional convertible issued on the same premium, as the share price appreciation called for to make conversion optimal is greater. For example in a premium redemption structure with 110% redemption price, parity needs to be over 110%, compared with 100% in a conventional structure, before a holder would elect to convert rather than opt for redemption at maturity. Because of the lower probability of conversion these bonds tend to be nearer the ‘debt’ end of the debt/equity spectrum.

Put features

Zero coupon and OID convertibles often carry investor put options, with the put price set to the accreted price of the security on put date.

Call protection

It is quite common to find cases in which an unconditional call occurs at the same time and price as an investor put. In these cases the ‘matching-put-and-call’ date can be looked upon as the effective maturity date of the security, since the circumstances in which it would not be rational for holders to exercise their put are likely to be those in which the issuer would be inclined to exercise the call. In cases where the zero coupon convertible does survive a matching-put-and-call it tends to trade close to parity or be voluntarily converted by holders on income grounds.

Contingent features

A contingent conversion feature limits the investor’s ability to convert. It will specify a share price below which conversion is not permissible except in certain circumstances. Though the incorporation of such a clause in the terms of an issue is naturally against the investor’s interest, it is often immaterial as the bondholder would be unlikely to wish to convert in those scenarios in which he may not. Contingent interest payment features allow the payment of a small amount of interest to the convertible bondholder in certain circumstances, such as share price weakness at some period in the future.

From the investor’s perspective

Zero coupon and OID bonds are frequently more defensive instruments than coupon convertibles because the market price often stands below the redemption price. Figures 8, 9 and 10 compare a notional zero coupon premium redemption convertible and a notional conventional convertible.

Figure 8. Zero Coupon Bond with Premium Redemption Structure

Issue Price	100
Coupon	0%
Redemption Value	120
Maturity	4 years
YTM	4.66%
Conversion Ratio	14 shares per bond

Source: Citigroup Convertibles.

Figure 9. Vanilla Convertible Structure

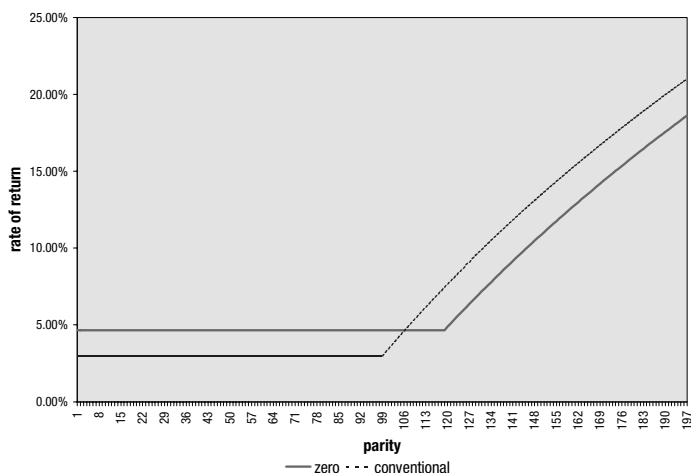
Issue Price	100
Coupon	3%
Redemption Value	100
Maturity	4 years
YTM	3%
Conversion Ratio	14 shares per bond

Source: Citigroup Convertibles.

The coupon-bond has a higher probability of being converted at maturity than the premium redemption bond. If at maturity parity is 106, the coupon-bond will be converted into the underlying shares as the redemption price of 100 is below parity. The zero coupon bond, on the other hand, will be redeemed at 120 in this instance. The effective economic conversion price, the price at which the investor is indifferent between redemption and conversion, is higher in the premium redemption structure than in the cash coupon convertible.

Figure 10 shows the payoff of the two structures. The defensive character of the premium redemption structure is illustrated by the fact that it does better than the conventional convertible in negative scenarios and less well when the share price is strong.

Figure 10. Rate of Return to Maturity



Source: Citigroup.

A vanilla convertible compared with a bond + warrant

Though a convertible can be thought of as a combination of a straight bond and a call option, there is a difference between buying a convertible and buying a 'bond cum warrant' issue (a bond issued with a long maturity call option attached). A bond cum warrant package can usually be stripped into its components and traded separately in the secondary market. In a convertible the call is embedded in the security. Investors in a bond cum warrant can typically exercise their option by subscribing cash instead of bonds leaving the bond portion in existence. For a convertible, this is not the case.

Mandatory convertibles

DECs

DECs stands for '*dividend enhanced common stock*' or, in the case of exchangeable DECs, '*debt exchangeable for common stock*'. A DECs is typically structured as a convertible preferred share that pays a fixed quarterly dividend. The yield at issue is higher than the current yield of the underlying common stock. A DECs will automatically convert into the underlying ordinary shares of the issuer at maturity; in some cases it can also be converted prior to maturity.

As with a traditional convertible security, the coupon and premium are set at issue. The term '*conversion premium*' has a different meaning for a DECs, however. The conversion ratio of a DECs varies according to the level of the underlying share price at the date of conversion. The 'true' premium is a moving target. By convention, the premium quoted is that based on the lowest conversion ratio (or highest conversion price) that could apply. DECs typically have a maturity of three to four years.

Adjusted conversion ratio

The number of shares received upon conversion depends on the price of the underlying shares at maturity or in cases where early conversion is possible, at the time of conversion. The minimum conversion price is normally the share price at issue and the maximum conversion price is typically some 25% higher. The issue price of a DECs is sometimes the price of the share at issue though many now have some other issue price such as 50 or 100 (in the appropriate currency).

The example below assumes the issue price is 100. The share price at issue is denoted by S_a and the share price at conversion by S_b . The maximum conversion price is denoted by S_c .

The conversion price will be set according to the following rule:

- When the underlying share price is at or below the share price at issue (S_a), the conversion ratio is $100/S_a$.
- As the share price moves between the price at issue and the maximum conversion price, the ratio is set such that the value of the shares delivered is constant. The conversion ratio is $100/S_b$
- Above the maximum conversion price, the conversion ratio no longer falls and is fixed at the minimum level set at issuance. The conversion ratio is $100/S_c$.

Figure 11. DEC: number and value of shares received

Stock Price at conversion	Number of Shares Received	Value of the Shares Received
$S_b < S_a$	$100/S_a$	$S_b \cdot 100/S_a$
$S_a < S_b < S_c$	$100/S_b$	$S_b \cdot 100/S_b = 100$
$S_b > S_c$	$100/S_c$	$S_b \cdot 100/S_c$

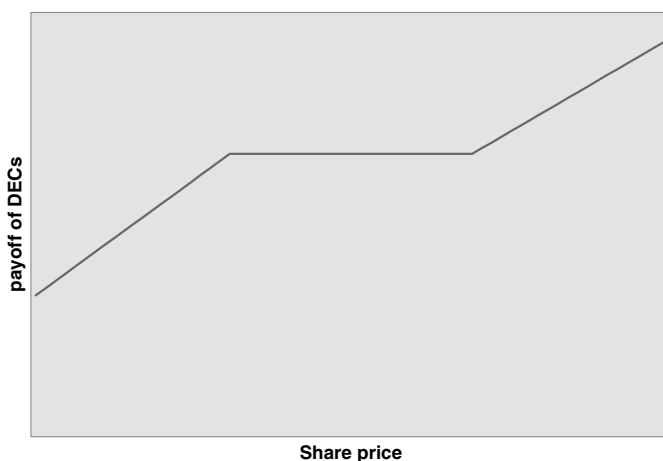
Assumed issue price: 100

Denotation as above

Source: Citigroup Convertibles.

DECs payoff

Figure 12. DEC: Payoff Chart



Source: Citigroup Convertibles.

Many investors perceive a DEC as a form of ordinary share where an element of equity participation in particularly strong share price scenarios is sacrificed in exchange for a higher yield.

A DEC is usually non-callable for life.

Exchangeables

Exchangeables are bonds issued by one company that exchange into the shares of a different company. They can be issued in both mandatory and non-mandatory form.

From the issuer's perspective

Exchangeable bonds can permit a company to dispose of or monetise a shareholding in another company. Not surprisingly, the justification for issuance bears a strong resemblance to that for any convertible. If the issue remains unconverted, the issuer will have secured low cost funding. If it gets converted, the issuer will have succeeded in disposing of the stake at a premium to the price that ruled at the time of issue. In many tax jurisdictions the disposal of a stake by means of an exchangeable bond enables the issuer to defer the crystallisation of tax liabilities on the sale.

One advantage of the exchangeable over a pure placement is that the issuer continued to enjoy voting rights.

Governments have also considered exchangeables as an effective way of monetising shareholdings as part of privatisation programmes. The exchangeable route can permit progress on a privatisation plan to be made even at times when the equity market would not be receptive to a pure equity offering.

From an investor's perspective

Exchangeable bonds present different valuation challenges to normal convertibles. Three additional issues are of importance: correlation, ring fencing, and forewarning.

Correlation

It is clear that if the issuer and underlying companies are one hundred percent correlated, there is no difference between the exchangeable bond and the convertible. Where there is a degree of independence there are more default scenarios in that either the issuer or the underlying company can default. In the event of underlying entity default the payoff is *bond floor* compared with *recovery rate* in the convertible.

Ring fencing

The payoff of the structure in the case of issuer default, no ring fencing and no forewarning is the same as that of a convertible (*recovery rate*). In the case of ring-fenced convertibles the payoff on default of the issuer is *max (parity, recovery rate)*.

Forewarning

With perfect forewarning of issuer default, the payoff of the non-ring-fenced convertible is the same as that of the ring-fenced convertible *max (parity, recovery rate)*. As stated above, the payoff in the case of no forewarning is *recovery rate*.

It has been found possible to adapt many convertible models to incorporate most of these features. In general zero forewarning is implicitly assumed, however. This penalises the non-ring-fenced deals in a manner that may be inappropriate.

Reset features

Reset features allow for a reduction in the conversion price of a convertible in the event of share price depreciation.

Typically a reset clause will set down the reset date or dates, a maximum conversion price (cap), a minimum conversion price (floor) and a scaling factor. The new conversion price is typically based on the average share price level over a specified period shortly before the reset date. The reference level is multiplied by the scaling factor (often 1) to generate a possible new conversion price. If this number is between the floor and the cap it is adopted as the new conversion price, otherwise the floor or the cap is adopted.

Since share price weakness can increase the expected conversion ratio, it can sometimes provoke hedged investors to sell shares, reversing the normal process whereby rehedge activity tends to bring buyers on share price falls and sellers on share price rises. This 'negative gamma' effect can be destabilising, with share price falls and hedge fund selling feeding each other in a vicious spiral.

The DEC's structure mentioned above falls under the general heading of a reset convertible. It is a mandatory reset convertible with one reset at maturity.

Floating rate convertibles

Floating rate structures are not common in Europe. Having a floating rate coupon does have some merits however, particularly for issuers who prefer variable rate financing and would otherwise need to use the swaps market to convert a convertible bond's fixed coupon into floating.

Because the life span of a convertible is less predictable than that of other bonds, specifying such a swap can be cumbersome. Some issuers have a natural inclination to prefer variable funding due to the nature of their business. For others it will be expectations that drive the preference. For example those who believe rates will be lower for longer than is implied by the yield curve would save on expected debt servicing costs by issuing an FRN.

From the investor's point of view the major disadvantage of floating rate convertible bonds is that valuation is rendered more complicated. In some cases it may be possible to identify an equivalent fixed coupon bond for valuation purposes. Where there are calls, however, there is no accurate way of doing this; it is necessary to create a table of the future expected coupons as implied by the term structure of interest rates. It may be necessary to model a perpetual FRN as one with a finite date, though using par as the 'redemption price' in such circumstances is arbitrary. There is a case for using the expected value of a perpetual FRN as the redemption price.

A feature of FRNs is that the bond floor tends to be much more steady over time than in the case of a fixed coupon convertible. This means that a bond set up to capture the nature of a floating rate convertibles through a 'dummy' fixed coupon bond will not produce a good estimate of the rho (DV01) of the security.

Pitfalls and Protection

- The prospectus
- Special dividends
- Takeovers
- Dilutive events
- Income entitlement
- Clean up calls
- Other prospectus issues

The prospectus

Convertibles are vulnerable to certain hazards in addition to those of straight bonds. These fall into two general categories:

- Events that would tend to lower the fair premium of the convertible. One example of this would be a takeover for cash that leaves the bondholder with rights of conversion into cash, or near cash.
- Events, such as rights issues, which have a dilutive effect on the underlying share price.

The prospectus will also provide information on the dividend and coupon entitlements of a convertible bond.

Special dividends

Unanticipated distributions such as special dividends, will put convertible bondholders in an unfavourable situation relative to shareholders. In some cases, convertible bondholders may even be forced into early conversion in order to participate in the distribution.

Most modern convertible prospectuses contain language designed to protect holders against capital distributions. The definition of a special dividend in Europe generally falls into one of three categories:

- In some issues the prospectus defines a capital distribution as dividends in excess of a given payment per share. This is frequently a certain percentage of the previous year's payout, or the highest annual payout over a certain number of prior years.
- In other cases payouts that push the dividend yield beyond a specified limit are defined as capital distributions (with variations for whether a share price average is used or share price at time of payment). Once the payment has satisfied the condition, the dilution compensation sometimes looks back to the previous year's payout, rather than taking the excess over the yield threshold. In French convertibles it is common to see the base level for working out the level of compensation also determined by a yield. For example one could have a dividend in excess of 5% triggering compensation, but the compensation itself being calculated on the excess over 3.5%.
- In some cases there is 'distributable profits' rule, which states that only distributions in excess of earning per share are capital distributions. This category can be particularly detrimental to bondholders, especially if the current payout ratio is not high. Furthermore the 'distributable profits' rule is sometimes cumulative; in essence permitting issuers to pay out retained earnings since the launch of the convertible without triggering a change in the conversion terms.

Anti-dilution provisions

Anti-dilution provisions cover a range of situations that involve dilution of the underlying share price. The provisions will stipulate the situations in which convertible holders are compensated for a dilutive event.

Typically this will involve an improvement in the conversion terms according to a formula. The intention is generally to increase the conversion ratio such that parity (that is the equity value of the convertible) is left unchanged, after accounting for the fall in the share price resulting from the dilutive event.

Buy backs

Some convertibles provide no protection against share buy backs in spite of the fact that a buy back above the current market price is likely to depress the share price after completion. Others specify an adjustment if the buy back level is materially (often 5% or more) above the then current share price.

Stock-split

Conversion ratios are almost always adjusted in case of the most straightforward of dilutive events, the stock-split. If the stock-split were on a one-for-one basis for example, the conversion ratio would be doubled.

Merger or takeover

In the case of a merger or takeover of the issuing company by another company, the treatment of the convertible bondholder will depend on the conditions of the prospectus or trust deed²⁶. In some cases there is no protection.

In general, where there are provisions for a merger or takeover event, there is a distinction in treatment of a bid (mainly) for cash and one that is (mainly) for shares. In an all-share bid by a quoted company, it is generally considered reasonable for the rights of conversion to be transferred into the shares of the new entity on a basis equivalent to the terms of the deal (the '*see-through*' basis). The bidder would also assume responsibility for the coupons and repayment of the convertible, if necessary. There is no presumption as to whether such a turn of events will be advantageous for convertible holders. It will depend, inter alia, on the volatility of the new company relative to that of the old, the dividend policies of the new company and its credit status.

²⁶ In most jurisdictions laws and take-over codes provide additional protection. For example in the UK an offeror is required to make an appropriate offer for any convertible bond of the target company (Takeover Code rule15).

The danger in the event of a cash bid is that the exchange property of the convertible becomes cash or near-cash. In this situation, the lack of potential movement in the value of the exchange property would lower or extinguish the premium. A number of devices can be applied to address this situation.

The three most common forms of takeover protection are a ratchet (discrete or continuous), an adjustment based on recent trading levels of share and convertible and cash compensation formulae.

Ratchet

Most ratchets specify that if a takeover occurs in a given time frame, improved conversion terms will be available for a limited period, such as 60 days. Typically there will be a schedule of deteriorating adjustments for successive time periods. A number of European bonds have a continuous adjustment formula. It may start with a conversion ratio change equal to the original issue premium and diminish to zero each day over a period of three years, for example.

Premium based adjustment

Rather than refer back to the premium at issue, some bonds provide for compensation based on the recent ratio of the trading prices of the share and the convertible. The period of time looked at is usually about 30 trading days. In dual currency bonds it is important to pay attention to the exact adjustment formula. Sometimes the adjustment, quite logically, brings parity on the new terms up to the market price of the convertible (for the calculation period). In some cases however 'parity' is implicitly calculated by reference to the share price and conversion price rather than share price and the conversion ratio. In dual currency bonds the adjustment can be surprisingly generous or ungenerous depending on the behaviour of the foreign exchange rate since issue.

Cash Compensation formula

Some European convertibles employ a formula that seeks to compensate holders according a number of criteria. The length of time since issue, the degree of cash in the takeover, parity at issue and parity at takeover are all taken into account. This style of adjustment is sometime referred to as a 'k squared' formula as the square of the ratio of current parity and issue parity is frequently an element in the formula.

Investor's put

Approximately half the outstanding European convertibles give bond holders a put at par, or in the case of accreting bonds at accreted price, in the event of a take-over or merger.

Income entitlement

In the European convertible market, a measure of standardisation now exists on the question of dividend entitlement. The general rule is that accrued interest is not paid on a convertible on conversion and the shares delivered on conversion will rank *pari passu* with existing shares. There are, however, a number of exceptions to this rule. In particular, if holders are 'forced' to convert by the sending out of a call notice, there is often language giving protection against the scenario in which a long period has passed since a convertible coupon has been paid yet converting bondholders have just missed a dividend payment. Other bonds permit delivery of ordinary shares that are ineligible for one or more dividends. For example, the entitlement to dividends may be based on the financial year in which conversion takes place. A phenomenon frequently encountered in French companies with a calendar financial year is that one needs to convert by December 31 of one year to get shares that are eligible for dividends paid in May or June of the following year. An extra coupon will often be secured by converting the following January but apart from missing a dividend the expense and inconvenience of converting into shares that cannot be delivered against a short position in normal shares must also be taken into account.

Sometimes the 'financial year' rule applies if new shares are delivered but the 'record date' rule applies if existing shares are received. This means the analyst needs to make a judgment on the relative probability of new or old shares being delivered.

Clean-up calls

If only a small percentage of the original issue size of a bond remains outstanding, the issuer may be entitled to call any remaining bonds. The percentage will be often set at around 10% of the issue size.

Other prospectus issues

Exact details of call and put conditions are worthy of scrutiny.

Call conditions in soft call bonds

In soft call bonds the trigger permitting a call is sometimes harder to achieve than is supposed. In some cases a bond is thought to be callable on a certain date (subject to condition) but the prospectus dictates that this is the earliest that a notice can be dispatched. With a typical minimum notice period of 30 days the effect is to make the bond non-callable for an extra month. In other cases the trigger is based on the level of the share price over a number of days and this reckoning cannot start until the announced call date. This pushes the first day on which a call can become effective even further forward.

The most important considerations relate to the precise conditions that need to be fulfilled, however. In most cases the trigger is more complex and harder to achieve than a 'one touch' share price condition. There are two principle soft call rule types; those that state that the share must be above a certain threshold for a certain number of days and those that specify that the average share price in a given period must exceed the threshold. Within the 'days above' category there is a further subdivision between those that require the days to be consecutive and those that allow the share to fall below the boundary provided it is above it on a certain number of days within a specified period. By far the most demanding condition is that which requires the share price to remain above a certain level for considerable number (say 30) of consecutive days.

Trigger currency

It is possible to make a serious valuation error (in either direction) in cross-currency convertibles (those in which the currency of the share and that of the bond are different) by failing to note whether the call trigger is specified in terms of the domestic share price or the value of the share expressed in terms of the bond currency. A trigger in terms of bond currency is the more logical as an investor will know the level of parity that corresponds to the call condition. There are, however, many cases in which it is the share price in domestic currency terms that counts.

Put dates

It is important to look closely at the precise wording of an investor's put right. Where a considerable notice period is required and the put notice is not rescindable, the effective put date is earlier than a superficial inspection might suggest, detracting from the fair value of the convertible.

Convertible Pricing Models

- **Recursive techniques**
- **Two factor models**
- **Approximations using European options**

Recursive techniques

Most convertible pricing models employ a binomial tree or similar approach. This uses a recursive procedure in a manner similar to that employed in valuing American options. The approach can deal with most of the common features found in convertibles — it can capture not only the value of the early conversion right, but also calls, puts and other characteristics.

As with other options models, binomial trees deal with probabilities. It is easy to misunderstand the nature of these probabilities, however. They do not relate to the chance of certain events occurring in the real world, but are set such that the 'expected price' of the share at any given date is constrained to equal the theoretical forward price at that date; the forward price being determined by arbitrage principles not expectations.

The tree works by dividing the life of the bond into specific points in time, with a range of convertible prices considered for each. Apart from the final (maturity date) range each price is calculated from two prices from the chronologically succeeding range. The relationship between the price and the two prices from which it is composed is given by the volatility of the share and a drift factor.

One starts by associating a value for the convertible with each share price level at the maturity date of the bond. In general, for share price levels above the conversion price the value is parity and for share price levels below the conversion price it is 100%. One now simply works backward through the tree.

The next stage will be one period nearer the present day, with the period length being determined by the term of the convertible and the number of steps chosen. A new distribution for this stage is established and a value is assigned to a range of outcomes. In some cases the value for the convertible associated with each node calculated in this manner will need to be replaced if some feature of the bond makes this appropriate. For example if the bond is puttable at 100 on a certain date, all observations under 100 at this date will simply be replaced with 100.

This process continues until one arrives at the present day with a fair value for the bond.

Formulae for the share price outcomes at maturity, the probability of an up, the probability of a down and the drift rate per period are based on simple algebra and can be found in options text books.

Two factor models

One limitation of the binomial tree approach is that it allows the share price to follow a stochastic path, but treats all other inputs, such as bond yields, dividends and FX rates as being either predetermined or dependant on the share price. Of these, the lack of a random element in the interest rate assumption is a particularly significant limitation.

Models have been developed to allow more variables to be modelled, for example incorporating stochastic bond yields and bond yields that are correlated to equity prices. Generally they employ recursive procedures similar to those of simple binomial trees. The number of factors in a model refers to how many variables are treated as stochastic. A one-factor model varies just underlying stock prices, whereas a two-factor model takes account of volatility in interest rates (for example) as well.

Approximations using European options

From an equity investor's perspective

We have established that the value of a convertible to an equity investor can be summarised as:

$$\text{Convertible Price} = \text{Parity} + \text{Put Option Value} + \text{Value of Income Advantage}$$

In simple, non-callable convertibles, which are unlikely to be converted early one can estimate the theoretical value by establishing valuations for the components:

- Parity is simply the conversion ratio multiplied by the current stock price expressed in the bond currency.
- The put option value can be calculated using the Black Scholes model. The put is interpreted as the right to sell a parity number of shares at the redemption price.
- The income advantage is calculated as the present value of the expected income difference.

By adopting this approach, equity investors can obtain a very clear impression of what they are paying for when they buy a convertible at a premium. However, the approach fails when there is a good chance of voluntary early conversion or when there is a period in which the bond is callable. When the call is provisional, even approximations in which the first call date is treated as the effective horizon are unreliable²⁷.

From the bond investor's perspective

The value of a convertible to a fixed income investor could be summarised as:

$$\text{Convertible Price} = \text{Bond Floor} + \text{Call Option Value}$$

²⁷ This approach highlights a shortcoming of the conventional approach. A soft call bond can be seen as (long a bond, long a call option and short another complex option). However because of the high strike of the option against the holder it may be appropriate to employ a lower volatility for it.. Most models use just one volatility input and thus tend to overvalue the option against the holder of the convertible.

The theoretical value of a non-callable convertible can be calculated by valuing these components:

- The bond floor value is calculated as the present value of the future coupon payments and the cash redemption value.
- The call option value can be calculated using the Black Scholes model set for a vanilla European call, exercisable at maturity.

As with the 'share plus put' calculation, the major advantage of this approach is its transparency for investors. The problem is that the approach is only theoretically correct if the bond happens to be non-convertible and non-callable for life. The extent of the potential error depends on how different the terms of the convertible in question are to this basic description. If the bond is convertible at any time, the approach is deficient in that it ascribes zero value to the early exercise option. However where there is a significant income advantage in the convertible, the early exercise right is of little or no value anyway. In the case of bonds with an unconditional call feature, the use of approximations can be relatively successful. Where there is a provisional call period, however, the application of the European option methodology breaks down.

A note on the use of risky rates

Many convertibles models employ a 'risky' yield curve throughout. To those taught to believe in the sanctity of risk neutral valuation this can be hard to come to terms with.

The rationale for pricing with credit spreads rests on a particular model of default. Each company is seen as having a 'hazard rate' that determines its chance of defaulting on any day. Default is typically seen as striking, with the predetermined probability²⁸, 'out of the blue'²⁹.

Once this 'jump-to-default' possibility is accepted, it becomes clear that call options on risky shares are more valuable than those on otherwise identical riskless shares. Consider the owner of a call option with market price 10 where the share price is 100 and the delta is 50%.

²⁸ The time to default is modelled as an exponentially distributed random variable. The probability of default over $[t, t+dt]$ given that there was no default up to t is then given by a constant (the hazard rate) times dt .

²⁹ This model of default is sometime referred to as 'blue sky default'.

If the share price were to go to zero he would lose 10 points on his option but make 50 points on his hedge, securing a 'windfall' of 40. Clearly the more likely the incidence of default the more valuable the option.

The occasional windfall made on default can be turned into regular incremental returns by selling default protection³⁰.

Appendix

Introduction

This appendix contains additional information on yield to maturity (YTM), bond floor, delta, gamma, vega, rho, theta, and fugit.

Yield to maturity (YTM)

YTM can only be calculated through a process of iteration. The formula below shows the relationship between the bond price, the (annual) YTM, the coupon and the redemption amount in a simplified case where there are an exact number of years until maturity of the bond and coupons are paid annually.

$$\text{Convertible Price} = \sum_{i=1}^m \frac{C}{(1 + YTM)^i} + \frac{R}{(1 + YTM)^m}$$

Where:

m = years to maturity;

C = coupon; and

R = redemption value.

³⁰ To be more precise the replicating portfolio for a call option comprises a short position in the shares (on a delta) and a long position in corporate bonds of the issuer. This model of default suggests another solution to the problem of identifying the correct credit hedge in a convertible. Assuming zero recovery the profit on the hedge on default is parity*delta, so the credit hedge should be CB Price -parity*delta.

In the case of the *Chimera Convertible EUR 4% 2011*, the YTM of the issue is that which solves the equation:

$$€1,000 = \sum_{i=1}^5 \frac{€40}{(1 + YTM)^i} + \frac{€1,000}{(1 + YTM)^5}$$

Solving for YTM (through trial and error)

$$YTM = 4\%$$

Bond floor

One formula that can be used to calculate a convertible's bond floor bears a strong resemblance to the YTM formula above.

$$\text{Bond Floor} = \sum_{i=1}^m \frac{C}{(1 + d)^i} + \frac{R}{(1 + d)^m}$$

Where:

m = years to maturity;

C = coupon;

R = redemption value;

d = discount rate (the risk free rate + a credit spread)

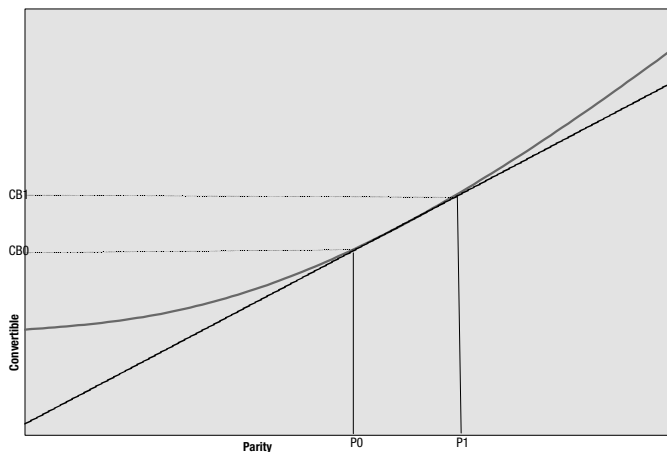
The formula applies where there is a whole number of years to maturity. A somewhat more complex formulation is required when there are fractional periods, though the principal of finding the discounted present value of the fixed payments is the same. When there are fractional periods, the value is 'dirty'. In order to quote it, we need to deduct the accrued interest.

The discount rate is calculated using the risk-free rate for the maturity of the bond, plus a credit spread that reflects the credit quality of the issuer. In Europe, it is common to adopt the swap rate (Libor) in the currency in which the convertible is denominated as the risk-free rate. It is necessary to make an assumption on the credit quality of the issuer.

Delta

Figure 13 displays the sensitivity of the price of the convertible to changes in parity. It shows that delta is the slope of the tangent drawn on the convertible price curve at the level of the current share price (parity).

Figure 13. Convertible Price Against Parity



Source: Citigroup.

Figure 13 shows how the tangent can be an accurate representation of convertible price movements only for small share price movements.

If parity moves from P_0 to P_1 , then

$$(P_1 - P_0) * \text{Delta} \approx CB_1 - CB_0$$

If we consider infinitesimal changes, d , then

$$d(\text{Parity}) * \text{Delta} = d(\text{Convertible Price})$$

$$\text{Delta} = d(\text{Convertible Price}) / d(\text{Parity})$$

Figure 14. Delta Against Parity



Source: Citigroup.

Gamma

Gamma is the rate of change of delta for movements in the underlying share price:

$$Gamma = d (Delta) / d (Parity)$$

Conventionally, gamma is expressed as the change in delta for a one-unit change in parity.

We can interpret:

Delta as the first partial derivative of the convertible price with respect to parity; it measures the slope of the tangent drawn on the convertible price curve at the current level of parity.

Gamma as the second partial derivative of the convertible price with respect to parity; it measures the degree of convexity of the convertible price curve at the current level of parity.

$$\text{Gamma} = d (\text{Delta}) / d (\text{Parity})$$

$$\text{Gamma} = d^2 C / d P^2$$

Where

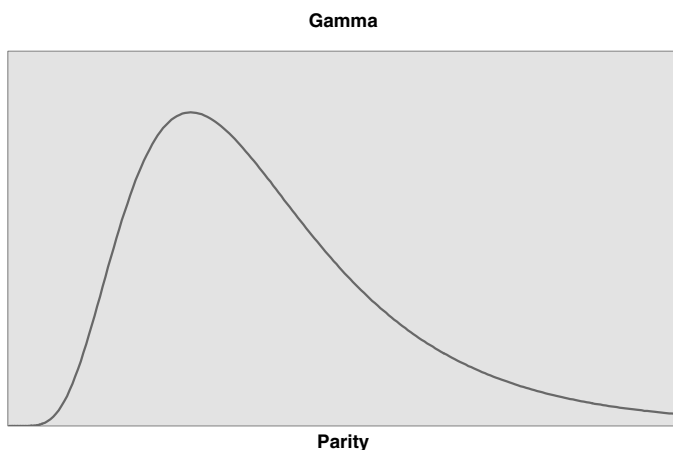
C = Convertible Price

And

P = Parity

For small stock price fluctuations, delta multiplied by the parity move can be a good approximation for changes in the convertible price. Adjusting for the effect of gamma is particularly important for larger share price movements, as the effect of convexity on the convertible price can be significant.

Figure 15. Gamma Against Parity



Source: Citigroup.

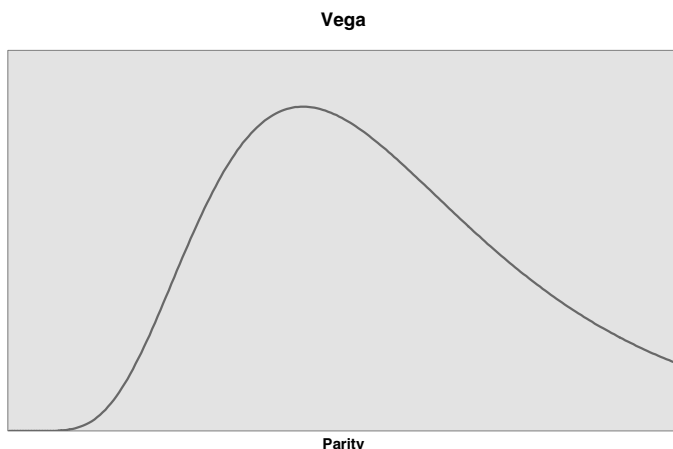
Gamma changes along the convertible price curve. Gamma is always positive in conventional convertibles, and it is at a maximum when the convertible is close to the money.

Vega

Vega is the sensitivity of the convertible price to changes in the volatility of the underlying stock. It can be expressed as follows:

$$Vega = d (\text{Convertible Price}) / d (\text{Stock Volatility})$$

Figure 16. Vega Against Parity



Source: Citigroup.

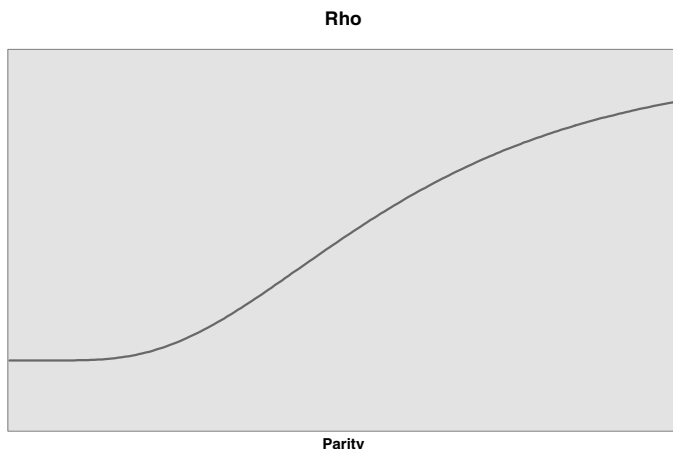
As can be seen from Figure 16, vega is always positive on a standard convertible and is greatest when the convertible is close to the money.

Rho

Rho measures the sensitivity of the convertible price to movements in interest rates. Rho can be expressed as follows:

$$Rho = d (\text{Convertible Price}) / d (\text{Interest Rate})$$

Figure 17. Rho Against Parity



Source: Citigroup.

Higher rates decrease the value of the bond floor but increase the value of the embedded option, however the former influence tends to predominate. Furthermore, as the share price falls the bond floor becomes an increasingly important component of a convertible's valuation and the sensitivity of the convertible price to changes in interest rates increases.

Theta

Theta is the change in convertible price with the passage of time. Conventionally, it is expressed as the change in the convertible price for the passage of one day, all other things being equal.

$$\text{Theta} = d(\text{convertible Price}) / d(\text{time})$$

As a convertible approaches final maturity, we see two opposite effects on the convertible price:

The time value of the embedded call option decreases and so the convertible price decreases; and



The bond floor trends towards the redemption value over time.

At-the-money and in the money convertibles will usually suffer from the first of these effects, with theta being a negative number. For out-of-the-money convertibles, the drag to redemption of the bond element can be the more potent influence.

Glossary

Accreted value

OID and premium redemption convertibles 'accrete' over their life from issue price to redemption price. The prospectus sometime lays down a straight line appreciation but more often one that maintains the bond on its issue redemption yield.

Accrued interest

This is the value of the accrued portion of the coupon on a convertible bond. Most Eurobonds adopt a 360 day convention.

American-style option

This type of option allows the holder to exercise into the underlying asset at any time during the life of the option.

Anti-dilution provisions

These provide for an adjustment in the conversion terms in the event of special stock dividends, stock splits or other corporate events that can result in the dilution of the underlying share price.

At-the-money

A convertible is said to be at-the-money if the current share price is close to the conversion price.

Balanced convertible

A balanced convertible is a convertible that trades at a price where it is neither a pure equity substitute nor trading on its bond floor, but is balanced between the two.

Binomial tree

A binomial tree option-pricing model estimates the theoretical value for an option. Adaptations of the approach are commonly applied to convertible bonds. They take account of events such as puts and calls that take place during the life of the instrument.

Black-Scholes option pricing model

The option pricing model derived by Fischer Black and Myron Scholes is used to estimate the theoretical fair value of European options based on a range of inputs and assumptions.

Bond floor (or investment value)

The bond floor is the value of the fixed income element of the convertible if rights of conversion are ignored.

Bond with/cum warrant

This is a straight bond issued with a long maturity call option attached. The bond and warrant can typically be traded separately in the secondary market.

Breakeven

The breakeven calculation for a convertible measures the time taken for the bond's income advantage to offset the conversion premium. It is a simple measure that takes no account of dividend growth projections or discounting for present value.

Call feature (or call option)

A call feature gives a convertible issuer the right to redeem a convertible bond prior to maturity at a price determined at issue. Holders of convertibles who receive a call notice will generally have time to exercise their rights of conversion before repayment takes place; thus a call option can frequently be interpreted as required early conversion. (see also 'trigger')

Clean price

The clean price is the price of a convertible bond quoted excluding accrued interest. Most convertibles are quoted this way.

Clean-up call

If an issuer is entitled to call any remaining bonds when a certain percentage of the bond issue has already been converted into shares, or otherwise extinguished, it is termed a 'clean-up call'. The percentage is often set at around 90% of the convertible issue size.

Contingent conversion

A contingent conversion feature makes a convertible investor's ability to convert contingent upon the share price attaining a specified level.

Contingent interest payment

Contingent interest payment features allow the payment of a small amount of interest to the convertible bondholder if the average market price of the convertible falls to a specified level in a specified time frame.

Conversion premium

See premium.

Conversion price

At issue, the conversion price is the price at which shares are effectively 'bought' upon conversion, if the convertible is purchased at the issue price. It is calculated by dividing the issue price of the bond by the conversion ratio. It is market convention to define conversion price at maturity for a single currency bond as the principal amount divided by the conversion ratio, even for even for bond with above par redemption.

Conversion ratio

The conversion ratio is the number of shares into which each bond can be converted.

Convertible preferreds

Convertibles issued in the form of preferred shares. They pay a fixed 'dividend', equivalent to a coupon, and are sometimes issued in perpetual form.

**Convertible price**

This is the price at which the convertible is traded in the market. It is generally quoted as a percentage of par.

Coupon

The coupon is the interest payment per bond. It is normally quoted as a percentage of the face value.

Credit spread

The credit spread is the spread over the swaps curve (or sometimes Government bond curve) at which the issuer is assumed able to issue a straight bond that is otherwise identical to the convertible.

Cross currency convertible

A convertible that is denominated in a different currency to that of the underlying shares.

Current (or running) yield

Current yield is the income per unit of currency invested. It is calculated by dividing the coupon by the current convertible price.

Cushion

The excess of parity over the call price that an issuer requires before issuing a call notice.

DECs

DECS stands for *Dividend Enhanced Convertible Securities or Debt Exchangeable for Common Stock*. DECs are mandatory convertibles, typically issued as preferred stock paying quarterly fixed dividends.

Delta

Delta is a measure of the sensitivity of the convertible bond price to share price movements. It is defined as the expected change in the convertible price for a small absolute change in parity.

Denomination

This is the minimum size in which the bond can be traded.

**Dirty price**

The dirty price is the clean price of a convertible bond plus its accrued interest. It is the actual price an investor will pay for a bond.

Dividend yield

The dividend yield is an indication of the income generated by each share. It is calculated by dividing the annual dividend per share by the share price.

European option

This type of option gives the holder the right to exercise only on the maturity date.

Exchangeable bond

This is a convertible bond issued by one company that can be converted into the shares of a different company.

Gamma

Gamma measures the sensitivity of the convertible bond's delta to share price movements. It is the change in delta for a one-point change in parity.

Greenshoe

The greenshoe is an over-allotment option that allows an underwriter to increase the number of bonds issued, typically by 10%-15%, when there is strong demand for an initial offering.

Hard call protection

A period of time during which the issuer may not call the bond from the investor.

Hedge ratio

A convertible bond's hedge ratio is also referred to as 'delta'. The hedge ratio shows the equity sensitivity of a convertible bond and enables an investor to calculate how many shares he would need to sell to hedge the equity exposure.

**Implied volatility**

Implied volatility is the convertible pricing model volatility input that brings the fair value of a convertible into line with its market price.

In-the-money

A convertible is said to be in-the-money if the current share price is greater than the conversion price.

Issue price

The issue price is the price at which convertible bonds are sold to investors at issue.

Mandatory convertible

This is a convertible in which the bondholder is obliged to convert into the underlying equity.

Maturity

The maturity date is the final redemption date of the bond.

Nominal value

This is the face value of the bond. It is often 1,000 of the relevant currency in the Euroconvertible market and ¥1,000,000 in the Japanese and Euroyen markets. The current price, issue price and redemption price of most convertibles are expressed as a percentage of the nominal value.

Original issue discount (OID)

A convertible issued at a discount to par is termed an original issue discount convertible.

Out-of-the-money

A convertible is said to be out-of-the-money if the current share price is below the conversion price.

Par

Par is the face value of a bond.

Parity

Parity is the market value of the shares into which the bond may be converted. It is calculated by multiplying the conversion ratio by the current share price expressed in bond currency terms. It is normally expressed as a percentage of a bond's nominal value.

PERCs

PERCs stands for *Preferred Equity Redemption Cumulative Stock*. PERCs are a mandatory convertible bond structure that caps upside participation in a stock's performance.

Premium

A convertible's premium is the percentage by which the market price of the convertible bond exceeds parity. It represents the extra cost an investor must pay to buy the shares a bond converts into via a convertible. It is calculated by subtracting parity from the convertible price and is expressed as a percentage of parity.

Premium redemption structure

This describes a convertible bond that is issued at par but redeems at a premium to par.

Put feature

A put gives investors the option to sell back the convertible bond to the issuer at a fixed price on a given date or dates.

Ratchet

In some convertibles there is a ratchet mechanism in which the conversion ratio is adjusted by a specified amount if a takeover takes place within a given time frame.

Redemption price

The redemption price is the price at which the issuer must redeem bonds at maturity.

Reset date

The date on which a change of conversion terms takes place on a reset convertible is termed the reset date.

Reset features

Reset features allow for a change in the conversion price of a convertible in the event of share price depreciation (downward reset) or appreciation (upward reset) on certain specified dates.

Reset floor

The limit below which the conversion price on a reset convertible cannot fall.

Reset period

In most reset convertibles, the share price upon which the new conversion price is based is calculated by reference to the average share price observed in a specified reset period.

Rho

Rho measures the sensitivity of the convertible price to movements in interest rates. It is expressed as the change in the convertible price for a one basis point move in interest rates (a parallel shift in the yield curve).

Risk premium

The risk premium is the difference between the convertible price and the bond floor expressed as a percentage of the bond floor.

Soft call (or provisional call)

This is a period of time during which the issuer may only call the bond if the share price has traded above a predetermined level for a set period of time.

Step-up coupon

This is where the coupon level increases at a future date. This can be a contingent event on for instance, a credit rating downgrade.

Theta

Theta is the change in the convertible value with the passage of time. It is expressed as the change in the convertible price for the passing of one day, other things being equal.

Trigger

The threshold above which a share must trade before the issuer can dispatch a call notice. It is normally expressed as a percentage of the conversion price. Thus a 130% trigger means the share price must trade, for a specified period, at 130% of the conversion price. With dual currency convertibles the trigger is sometimes expressed in relation to the share price expressed in domestic currency terms and sometimes in the currency of the bond.

Vega

Vega is the sensitivity of the convertible price to changes in the volatility of the underlying stock. Vega is the change in the fair value of the convertible for a one percentage point change in the assumption for stock volatility.

Volatility

Share price volatility is a measure of the dispersion of share price returns. It is defined as the annualised standard deviation of returns. The extent to which the underlying share price has fluctuated over a certain period determines the historical or observed volatility. The assumption for future share price volatility is an input for convertible valuation.

Yield advantage

The yield advantage is the difference between the current yield on the convertible bond and the stock dividend yield.

Yield to maturity

Yield to maturity (YTM) is the discount rate that equates the current market price of a straight bond to the present value of its future cash flows.



This communication has been prepared by a member of the Sales and Trading Department of Citigroup which distributes this communication by or through its locally authorized affiliates (collectively, the "Firm"). Sales and Trading personnel are not research analysts and the information in this communication is not intended to constitute "research" as that term is defined by applicable regulations. Compensation of Sales and Trading personnel includes consideration of the performance of this Department's activities. The views expressed herein may change without notice and may differ from those views expressed by other Firm personnel.

You should assume the following: The Firm may be the issuer of, or may trade as principal in, the financial instruments referred to in this communication or other related financial instruments. The author of this communication may have discussed the information contained herein with others within the Firm and the author and such other Firm personnel may have already acted on the basis of this information (including by trading for the Firm's proprietary accounts or communicating the information contained herein to other customers of the Firm). The Firm performs or seeks to perform investment banking and other services for the issuer of any such financial instruments. The Firm, the Firm's personnel (including those with whom the author may have consulted in the preparation of this communication), and other customers of the Firm may be long or short the financial instruments referred to herein, may have acquired such positions at prices and market conditions that are no longer available, and may have interests different or adverse to your interests.

This communication is provided for information and discussion purposes only. It does not constitute an offer or solicitation to purchase or sell any financial instruments. The information contained in this communication is based on generally available information and, although obtained from sources believed by the Firm to be reliable, its accuracy and completeness is not guaranteed. Certain personnel or business areas of the Firm may have access to or have acquired material non-public information that may have an impact (positive or negative) on the information contained herein, but that is not available to or known by the author of this communication.

Financial instruments denominated in a foreign currency are subject to exchange rate fluctuations, which may have an adverse effect on the price or value of an investment in such products. Investments in financial instruments carry significant risk, including the possible loss of the principal amount invested. Investors should obtain advice from their own tax, financial, legal and other advisors, and only make investment decisions on the basis of the investor's own objectives, experience and resources.

This communication is not intended to forecast or predict future events. Past performance is not a guarantee or indication of future results. Any prices provided herein (other than those that are identified as being historical) are indicative only and do not represent firm quotes as to either price or size. You should contact your local representative directly if you are interested in buying or selling any financial instrument, or pursuing any trading strategy, mentioned herein. No liability is accepted by the Firm for any loss (whether direct, indirect or consequential) that may arise from any use of the information contained herein or derived herefrom.

Although the Firm is affiliated with Citibank, N.A. (together with its subsidiaries and branches worldwide, "Citibank"), you should be aware that none of the other financial instruments mentioned in this communication (unless expressly stated otherwise) are (i) insured by the Federal Deposit Insurance Corporation or any other governmental authority, or (ii) deposits or other obligations of, or guaranteed by, Citibank or any other insured depository institution.

This communication contains data compilations, writings and information that are proprietary to the Firm and protected under copyright and other intellectual property laws, and may not be redistributed or otherwise transmitted by you to any other person for any purpose.

Copyright © Citigroup 2006. All Rights Reserved.

Front page image:

Photograph of replica of the Chimera of Arezzo statue, reproduced with the kind permission of the photographer, Ugo Bardi of Università degli Studi di Firenze.

